

Challenges And Mitigations

TEAM RUBIXCUBE · SIH 2026 · PS 26097

Challenges and mitigations — SIH 26097

Written 2026-09-29. Every item below is a problem this team actually hit while building PS 26097, and the thing we did about it. Each one carries an **evidence** line pointing at the file, commit or measurement that proves it happened, so any of them can be defended if a judge pushes. Nothing here is hypothetical risk.

Format matches the deck's challenge slide: a challenge, one line of context, four mitigations. There are **22**, deliberately more than a slide can hold — pick and cut.

If only three go on the slide, use these

#	Challenge	Why it is the strongest
1	Dialect speech is wrong three words in ten	It is the technical thesis of the whole project, it is measured, and the honest version beats every other team's claim
13	A recommendation she cannot enrol in manufactures the CAG's 41%	Ties our architecture directly to the scheme's own audited failure
10	A shared handset would have read one person's answers to another	Nobody else will have thought of it, and it is a disclosure by a State instrumentality

Runners-up if a fourth is wanted: **11** (nothing says where she lives) or **12** (a third of the official register is dead courses).

1. Dialect speech is wrong three words in ten — and we nearly claimed the opposite

Best published error rates: Bhojpuri **27.8**, Chhattisgarhi **27.4**, Magahi **30.4**, Rajasthani **41.8** WER. An early draft of our own pitch said "*there is no speech model for Bhojpuri*" — which is false, and Bhashini (funded by the same ministry judging us) carries two of the four.

Stop transcribing, start classifying. The unit of the system is a field chosen from a closed set, not a sentence. A 30%-wrong transcript still yields the right answer	Four-rung ladder, cheapest first: word list with phonetic match → number patterns → small-model classification into the fixed list → spoken read-back. ~70% of answers never reach the model
Publish the measurement, never the claim. The slide shows word-error-rate <i>next to</i> field accuracy. That gap is the design	Claim the error rate, never the absence. Both the decision log and the problem-statement doc carry a dated, signed correction of our own mistake

Evidence: docs/decisions.md (2026-09-25 entry, CORRECTED 2026-09-28) · docs/Utsav/research/02-tech-landscape.md §1.2 · ARTPARK-IISc SraVaani-1.0 model card · Vistaar/GramVaani WER table (arXiv:2305.15386).

2. Fuzzy matching fixes one answer and breaks another

Sound-alike matching is what absorbs dialect. It also decided that पता ≈ पापड़ and नहीं ≈ नर्वी — turning "no" into "9th class" in a field that gates eligibility.

Guard rails found by testing, not by taste: phonetic matching only for single words of 5+ sounds, close-spelling only at 6+ letters	A bare "जी" no longer means yes, and filler words ("का काम") no longer make two different phrases look alike
Every fix is a regression test in <code>ai/tests/test_extract.py</code> — the bug can never come back quietly	Re-measured after each fix: answer accuracy on simulated phone audio went 87% → 97% on the same 38 answers
Evidence: <code>docs/Prashant/engine/01-engine.md §3 · docs/Prashant/ivr/05-measurements.md</code> (three dated runs) · <code>commit b6cbb1c</code> .	

3. A transliteration bug made cross-script matching silently useless

The romanizer **dropped every Devanagari vowel**, so सिलाई never matched `silai` . The same map emitted `c` for the letter every transliteration writes `ch` . Nothing crashed; matching just quietly did less than we thought.

56 assertions in the core self-check caught it — a silent-degradation bug has no error message, so only an assertion finds it	The core has no dependencies and no I/O , so the identical matcher runs in the browser, in Deno and in Node — one behaviour, not three
Three more of the same family found in the same pass: a closure capturing a stale session and discarding answers, option ids accepted from the wrong state	A Capacitor plugin proxy returned from an async function had its <code>then</code> called natively — found only because the check ran on the device, not in a tab
Evidence: <code>commit 4f892d2</code> (bug list in the message) · <code>packages/core/test/selfcheck.ts</code> .	

4. The speech vendor's live API disagreed with its own documentation

Written from the docs, verified against the wire: `saarika` is deprecated for `saaras:v3/v4` , `bulbul:v2` for `v3` , and — the one that mattered — **the response carries no n-best at all**, so the ladder's agreement boost could never fire.

Second pass with <code>language_code=unknown</code> to obtain a genuine alternate hypothesis, so the ladder has something to agree with	Restricted to the four dialects , because the second pass doubles speech cost and buys nothing where a real model exists
Verify every vendor fact against the live API , never against memory or a blog post	A vendor outage must not end an interview: an empty transcript reads to the flow as a timeout, so the assistant asks again instead of hanging up
Evidence: <code>commit 7d37573</code> · <code>commit b827969</code> · <code>docs/Prashant/ivr/05-measurements.md</code> .	

5. The speech vendor rate-limits, and we found the ceiling rather than guessing it

Back-to-back requests returned **429 Too Many Requests on 50 of 228 answers**, plus six timeouts over two seconds. On a real district that is a cap on simultaneous calls, not a lab curiosity.

Every one of the 50 fell back to on-device Vosk and the call continued — the designed behaviour, proven under the failure it was designed for	Two providers behind one switch: error or >2 s → Vosk → keypad menu. No path ends in silence
Both providers measured on identical seeded audio , so the comparison is real: worst line, Vosk 36% WER / 68% answers vs Sarvam 11% / 89%	The limit is documented as found, with the run table printed , instead of a vendor SLA we would have to take on trust

Evidence: docs/Prashant/ivr/05-measurements.md (unpaced run table) · commit b22876e .

6. One box handled 25 calls, because speech-to-text ran inside the engine's global lock

Calls queued behind each other's transcription. Ten simultaneous callers were already at 865 ms p95 for a reason that had nothing to do with speech and everything to do with a lock.

Transcribe before taking the store lock. After the fix, 25 simultaneous calls all finish, p95 silence 843 ms	A load tool that runs N full interviews from fresh numbers and counts only calls that reach the result — not calls that connected
The ceiling is stated, not hidden: ~25 per box with on-box speech. Past 35, turns hit the 1.5 s timeout	Degradation is graceful: the caller hears an apology and their progress is saved; cloud speech moves the load off the box entirely

Evidence: commit bbd87e1 · docs/Prashant/ivr/05-measurements.md (load table).

7. On a phone call, 1.8 seconds of silence is the whole product

Past roughly two seconds a caller assumes the line is dead. Speech-to-text alone can eat 800 ms of that, and an AI call on the hot path eats the rest.

Every fixed prompt is a pre-rendered file held in memory. No text-to-speech and no disk read on the hot path — about 80% of turns	Word list before AI: ~70% of answers resolve in ~5 ms. The model is budgeted for the uncommon path only
A 300 ms "हम्म..." plays at 700 ms , so a slow turn still sounds like a person thinking rather than a dropped call	One region, end to end (Mumbai). Measured 339–628 ms per turn against the 1,800 ms budget; the deployed speech function answers in 228–322 ms, faster than the laptop

Evidence: docs/Prashant/ivr/04-optimization.md §2 · 05-measurements.md (latency table) · commit 004b576 .

8. No voice-detection setting wins both noise and quiet speakers

At 10 dB signal-to-noise, aggressiveness 2 misses speech; aggressiveness 3 catches it and then loses quiet speakers. There is no value that is right for both a market and a shy 55-year-old.

Tested across 30 / 20 / 15 / 10 dB before choosing, and confirmed noise alone is never mistaken for an answer

Kept the setting that protects the quiet speaker, and let the loud-environment case fall to a nudge and then the keypad

Thresholds are a config knob, not a constant — they are meant to be retuned on real village recordings, which synthetic noise cannot stand in for

Shorter waits on noisy lines were tuned afterwards from real call feedback, not from the lab number

Evidence: docs/Prashant/ivr/README.md · progress log (step 4) · commits 61e054e , 566d6e7 .

9. We had no village audio, so we built the village line

We could not record in a district. Testing a rural IVR on clean laptop audio would have proved nothing, and every number from it would have been a lie by omission.

A phone-line simulator in the standard library: 300–3,400 Hz band limit, G.711 mu-law, noise at a set SNR, 5% lost 20 ms packets

Reproducible seeds, so a change is measured against the same degradation every time and an improvement is real

A browser softphone that speaks the real Exotel protocol — a spoken interview testable with a real mic and no telephony account

Labelled honestly everywhere: these are synthetic voices on a simulated line, *not* dialect audio. The measurements file says so in its first paragraph

Evidence: channels/ivr/tools/phone_line.py · commits b6cbb1c , e56bc5f · docs/Prashant/ivr/05-measurements.md (header note) · docs/decisions.md (caveat).

10. A shared handset would have read one person's answers to another

Keying the resume on a phone number meant a brother redialling would hear his sister's education, income and **mobility/disability disclosure** read back aloud — an unauthorised disclosure by a State instrumentality — and his answers would write into her record.

The phone number becomes a lookup index, not an identity. Any number of people share one handset; nulls stay distinct, so a phone-less beneficiary is a valid person

Resume is forward-only: it asks the next unanswered question and reveals nothing already given

The PIN gates only the two actions that disclose — full read-back and the saved recommendation — never entry. A forgotten PIN must not lock somebody out of a welfare line

The name gate was rejected as insufficient: a relative can guess a name. Two PIN failures start a new person on the same handset

Evidence: docs/Utsav/research/05-technical-spec.md §9 BLOCKER 1 · commits b856d68 , e6c4fcd , 90a4904 .

11. Nothing in the seven mandated fields says where she lives

The problem statement names seven fields and none is an address. A phone prefix does not resolve to an Indian district and the national register carries no geography — so the district demand report, which is our entire officer-side answer, had no GROUP BY .

Q0 — village/block, asked before consent with the language pick, resolved by the same word-list-and-phonetics matcher already built for trades	Presented as registration metadata, not an eighth question , so "seven fields, verbatim, in order" stays literally true for an evaluator counting them
A spoken village is taken as stated, with the official code left NULL. Forcing it into a twelve-block hard-coded list made Q0 never offer its confirmation at all	Free on kiosk and assisted mode — the device already knows its own block, so the question is skipped

Evidence: docs/Utsav/research/05-technical-spec.md §9 BLOCKER 3 · commit 2900fef · docs/decisions.md (2026-09-26, Q0 is district-level until the LGD list is imported).

12. The official register is a third dead courses and a fifth mis-mapped

The national register has **2,814 rows — of which 881 are expired**, 1,198 are valid at the levels our beneficiaries can enter. Underneath it, the regulator's own audit found **156 of 2,157 qualifications mis-mapped and 256 unmappable**.

"2,814" is never quoted as recommendable. The number that goes in front of a judge is the number that survives the filters	Nothing is invented: fields the official source does not provide are stored NULL. A wrong course code sends a real person to a centre that will not admit her
The snapshot is sha256-pinned and that hash is recorded on every recommendation , along with the version of the weights that produced it	Expiry is filtered in the query, not in an index. Postgres refused where valid_till >= current_date (42P17) and was right — such an index rots silently as rows expire past it

Evidence: commit 0aef648 (register counts) · commit 13fb8fe (42P17) · docs/decisions.md (qualification identifiers are imported, never invented).

13. A recommendation she cannot enrol in manufactures the CAG's 41%

The audited national picture is 56.14 lakh certified against 23.18 lakh placed. A system that suggests a course whose entry bar the person does not clear is producing that statistic one call at a time, politely.

A hard eligibility gate runs before ranking, never as a score — typed from the NSQF 2023 gazette table, including the schooling ↔ experience substitutions	Three buckets, not two: ELIGIBLE · NEAR-MISS with the exact gap ("one more year, or the Level 2 course first") · out. The near-miss is the problem statement's "skill gaps requiring intervention"
Experience only counts toward courses in the same trade as the work she actually did, and levels above 4 are never recommended	Concentration is priced into the ranking — at most two courses from one sector, and a spread penalty, because the audit found 40% of national certifications in ten job roles

Evidence: docs/Prashant/engine/01-engine.md §4–5 · packages/core eligibility gate · CAG Report No. 20 of 2025, Para 3.6 and Table 2.1(a).

14. We cited the wrong privacy rule, and the right one is narrower

We had built a guardian-consent branch on DPDP **Rule 10**. Rule 10 is children; **Rule 11** is persons with disability, and its test is a guardian appointed by a court, a designated authority or a local level committee. Firing that branch on every mobility disclosure is itself a dignity failure — most people with a physical constraint retain full capacity.

The branch now asks about decisional capacity , not about disability, and names the local level committee — which actually sits at district level and is the realistic route	The raw transcript is erased at confirmation. Its purpose is discharged, and it is 26–60% wrong anyway: near-zero evidentiary value, total disclosure risk
Two database CHECK constraints make it impossible to hold a raw transcript against a confirmed answer. The rule is enforced by the schema, not by a policy document	A privacy test greps the database and logs for ten-digit numbers, whole or partial PINs and audio columns before any real-user call

Evidence: docs/Utsav/research/05-technical-spec.md §9 MAJOR 4 and MAJOR 5 · commit a06e45c (privacy test) · commit 4f892d2 (CHECK constraints).

15. A vendor key inside an Android APK is one unzip away

The app first called the speech vendor directly. That ships the key to every handset, and sends caste-adjacent voice to a third party with nothing of ours in between.

Speech is proxied server-side , the key held as a platform secret. The app never holds a vendor credential	Only the public key ships in the app , which is exactly why every table carries a row-level security policy; the privileged key lives on the telephony VM alone
Mixed content turned back off. It had been allowed only so the WebView could reach localhost in development — and that wire carries a beneficiary's voice	The hashing pepper is generated once and never rotated casually , because rotating it orphans every stored phone hash

Evidence: commits b827969 , 004b576 , 7d37573 · docs/DEPLOY.md (secrets section).

16. The offline app's outbox was failing twice a batch and saying nothing

It looked idle. It was broken. Three bugs, and the third is why the first two survived so long.

The session was inserted before the beneficiary it points at — a foreign key, so every batch died on its first row	The phone-hash column was NOT NULL , which is wrong for the two channels built precisely for people with no phone: 51.6% of rural women 15+ own none
Both failures printed <code>[object Object]</code> . Database errors arrive as plain objects, so <code>.message</code> is undefined and the constraint code — the entire diagnosis — was thrown away	Errors now keep code, message, details and hint , a partial apply is treated as a failure instead of a success, and a failing sync no longer looks identical to an idle one
Evidence: <code>commit 2c5f2af</code> .	

17. An edge function cannot hold a phone call open

The whole serverless deployment was correct for everything except the one thing that pays for the channel: a four-minute bidirectional audio stream. Separately, a relative import that climbed out of the functions directory deployed cleanly and then failed at runtime.

One small always-on VM for telephony , in an Indian region, because the hot path is 20 ms audio frames and the voice must stay in Indian jurisdiction	The shared core is vendored in and its import specifiers rewritten — Deno resolves the literal path while Node wants the other extension. The copy is generated and re-synced on every deploy
The two traps that cost an afternoon are written down: Ampere capacity is routinely unavailable for days, and the stock Ubuntu image drops every port but SSH regardless of what the cloud console shows	The request/response pieces stayed serverless. Only the thing that genuinely needs a long-lived process got a VM
Evidence: <code>docs/DEPLOY.md</code> · <code>commit b827969</code> .	

18. A diagram generator cannot review its own diagrams

Our architecture diagrams are generated from the spec so the two cannot drift. But a validator can prove that text fits inside a box; it cannot see that an arrow points at nothing.

Render the generated file back to SVG and look at it. It immediately found a connector ending in empty space, which no assertion would have caught	Boxes are measured from their own text , so overflow is not representable — and the validator refuses to write the file if any of 41 boxes fails
Arrowheads are explicit paths, not markers , because a marker that silently fails to render would hide the exact class of bug the tool exists to find	The review found what a blind pass could not: the officer read as a fourth beneficiary channel, and had no arrows at all after login

Evidence: commits e6c4fcd , 4027217 , b856d68 .

19. The deck carried another problem statement's leftovers

Page 2 was titled "**KALA SETU**" — an artisan/handicraft project's name. Page 6 cited the Ministry of Textiles, MSME/ONDC catalogue onboarding, e-commerce exports and a paper on SME digital commerce. None has any connection to NSQF skilling for SC beneficiaries.

A full per-page fact-check against the primary sources, severity-tagged: blocker / unsupported claim / typo / upgrade

One hard fact error caught: 74.6% SC rural was a digit transposition of the Census's **76.4%**

Three phrases banned across all 26 slides: any claim of a portal API (the word "API" appears zero times in the 46-page guidelines), any dialect claim stated as fluency, and "verified" applied to a beneficiary — we confirm answers, we do not verify identity

Our strongest source now appears on the page that cites sources. The 41% figure had been printed without the audit it comes from

Evidence: docs/Utsav/deck-fixes.md (pages 2, 4, 5, 6 and the cross-deck section).

20. No field access — and the government already shipped something adjacent

We did no surveys and visited no districts. Meanwhile **Skill India Assistant** (MSDE + Meta + Sarvam, May 2025) is a live multilingual WhatsApp voice assistant that recommends NSQF courses. Somebody in that room knows this.

One rule for the pitch: nobody claims field work. Credibility comes from primary documents read in full — the CAG audit, the NSQF gazette, the PM-AJAY guidelines, the national register

The positioning is rehearsed, not improvised: SIA is national self-service discovery on a smartphone. This is feature-phone outreach, eligibility gating and district planning for people SIA cannot reach

We route to them rather than around them. The guidelines forbid overlap with PM-DAKSH by name, so PM-DAKSH routing is a stage in the recommender

The demo cohort is generated by the real engine, not hand-written: real profiles through the real ranker over the real catalogue, every row flagged as demo data on screen

Evidence: docs/Utsav/research/02-tech-landscape.md §7.1 · docs/decisions.md (route to PM-DAKSH and SIA) · web/app/src/lib/demo.ts (header comment).

21. Our own cost figure was wrong on its own terms

An early ₹5.60 per interview was wrong three separate ways, all checkable by a judge: inbound toll-free bills the *receiver* at a premium, the streaming feature is a paid add-on that was not in the budget, and the largest cost in the system had been left out entirely.

Missed call → callback on an ordinary number , not toll-free. The 2025 telecom amendment removes the regulatory reason toll-free was chosen, and saves 45–70% of the telephony leg	No always-on GPU . A self-hosted speech model costs ~₹4.29 lakh/year idle; against metered speech the break-even is ~47,700 interviews a month, far past a pilot
Every unverified vendor price carries a Δ and may not go on a slide without a written quote. Indian telecom vendors publish none of these rates	Rebuilt line by line to ~₹4.15 per interview , with the split stated — 37% telephony, 63% speech and AI — so the figure can be argued rather than asserted
Evidence: docs/Utsav/research/05-technical-spec.md §9 BLOCKER 2 and MAJOR 1 · docs/Prashant/ivr/01-master-plan.md §7 · 04-optimization.md §4.	

22. One design language cannot serve a beneficiary and an officer

Two full iterations were built and rejected. The first — cream on beige, serif display — is about 3:1 contrast and disappears in the direct sunlight of a doorstep interview. The second offered six affordances on a screen where the interview allows exactly one, with 28 px chips in a 44 px world.

Two registers over one token set , so the two halves cannot drift into two products: full-bleed one-decision screens for the doorstep, dense tables and charts for the officer	The beneficiary register holds itself to 7:1 contrast and ≥56 px targets, specifically for the sunlight and low-vision case
State is never colour alone , and the two accents are amber and teal — distinguishable under every common form of colour blindness, which green and red are not	Every beneficiary-facing string carries a prompt id and a replay control wired to the same audio file the phone channel plays. A string without one fails a lint test
Evidence: docs/DESIGN.md (three iterations, with the ratings and the reasons for rejection).	

What is still open, and said out loud

Kept here because a challenge slide that lists only solved problems reads as marketing, and because a judge who finds one of these unlisted will assume there are others.

Open item	Current honest position
The 30-utterance dialect measurement	The harness exists and is good; real recordings by real speakers do not. Until they do, dialect support is a claim, and we say so
Native-speaker prompts	Bhojpuri prompts are drafts read by a placeholder voice, marked DRAFT in the repo
The Perspective Plan's real column layout	We emit the district demand lines; nobody has yet opened the portal by hand to confirm the format it expects
Live opportunity data	Two or three hand-assembled pilot districts, every row carrying its source and date. No national feed is claimed
Outcome tracking	The table and the flow exist; today a human populates it. The follow-up call is designed, not built

